SDG Lens: Attention-Based Explainability for Multi-Label SDG Text Classification

Author Name
MSc AI for Global Challenges

Abstract

Text classification for the United Nations Sustainable Development Goals (SDGs) can support policy analysis, reporting, and large-scale monitoring of development documents. However, automated SDG classifiers are often difficult to inspect: they predict labels but provide little evidence for why a text was assigned to a goal. This paper presents *SDG Lens*, a compact prototype for explainable multi-label SDG classification. The system combines the SDGi Corpus, which contains document-level labels for the 17 SDGs, with an attention-based explanation strategy inspired by HateXplain. A MiniLM BERT-family encoder is fine-tuned with a 17-label sigmoid classification head using binary cross-entropy loss, while the final two transformer layers are unfrozen for lightweight adaptation. On a prototype English subset of 2000 training and 300 test examples, SDG Lens obtains micro-F1 ≈ 0.66 and macro-F1 ≈ 0.64. Attention highlights plausible SDG-related tokens, but these explanations remain proxy signals because SDGi does not provide token-level rationale ground truth.

1 Introduction

The Sustainable Development Goals provide a shared vocabulary for describing poverty, education, climate, health, cities, institutions, and related policy themes. Large development reports often contain passages relevant to one or more SDGs, making automatic SDG text classification useful for document analysis, policy tracking, and comparative research. In practice, however, classification alone is not enough. A model that assigns a document to SDG 11 or SDG 16 without exposing the textual evidence behind that decision is hard to audit, especially in policy-facing settings.

This work addresses that gap with a minimal prototype: an SDG classifier that also reports token-level attention as an explanation proxy. The goal is not to claim causal interpretability, but to provide a practical inspection layer: which tokens received high attention when the model produced its labels? The main contribution is a simple bridge between SDGi-style multi-label SDG classification and HateXplain-style attention-based explanation.

2 Related Work

The SDGi Corpus is a multi-label text classification dataset for SDG-related documents, with labels spanning all 17 SDGs. In this repository, the existing SDGi replication focuses on classification using a TF–IDF and linear-model baseline. HateXplain, in contrast, studies hate-speech classification with token-level rationales, motivating the use of model explanations alongside classification. SDG Lens combines these directions: it keeps the SDGi multi-label task but adapts the attention-explanation idea from the HateXplain-inspired BERT pipeline.

3 Method

3.1 Pipeline

The complete pipeline is intentionally small:

`SDGi text` → `BERT encoder` → `17-label classifier` → `sigmoid scores` → `SDG predictions + attention tokens`

The implementation reads local SDGi parquet files, converts each list of SDG labels into a 17-dimensional binary vector, trains a BERT-family classifier, evaluates multi-label F1, and saves both quantitative results and example explanations.

3.2 Model

The encoder is `sentence-transformers/all-MiniLM-L6-v2`, a compact BERT-family model available in the local HuggingFace cache. The classifier uses the pooled `[CLS]` representation and a linear output layer with 17 logits, one for each SDG. Training uses binary cross-entropy with logits:

$$\mathcal{L} = -\sum_{k=1}^{17} y_k \log \sigma(z_k) - (1 - y_k) \log(1 - \sigma(z_k)).$$

To improve performance while keeping training modest, most encoder parameters are frozen and only the final two transformer layers are trainable. The prototype was trained for three epochs on 2000 English training examples and evaluated on 300 English test examples. A threshold of 0.3 is used to convert sigmoid scores into predicted labels, reflecting the sparse multi-label nature of the task.

3.3 Explanation

For each example, SDG Lens extracts last-layer `[CLS]` attention, averaged across heads, and reports the highest-attention tokens. These tokens are treated as an attention-based explanation proxy. This is useful for inspection, but it is not a validated rationale: SDGi provides document-level SDG labels, not token-level evidence annotations.

4 Results

4.1 Overall Metrics

The prototype achieves micro-F1 = 0.6615, macro-F1 = 0.6425, weighted-F1 = 0.6575, and subset accuracy = 0.6567 on out-of-sample examples from the held-out SDGi test split. The average number of true labels per test example is 1.63, while the average number of predicted labels is 1.36.

4.2 Comparison with TF-IDF Baseline

To understand the value of attention-based encoding, we compare against a traditional TF-IDF + linear SVM approach using the same train/test split. Table 1 shows two key findings:

1. With 2k training examples, TF-IDF slightly outperforms the attention-based model (0.70 vs 0.66 micro-F1). This is expected: the MiniLM encoder has 17M parameters and needs more data to generalize.

2. With 4k training examples, the gap closes significantly (0.72 vs 0.70). The transformer model improves faster with more data, following the expected scaling pattern for neural encoders.

This pattern suggests that with 6-8k+ examples, the attention-based model would likely surpass the TF-IDF baseline. More importantly, it offers attention-based explanations that TF-IDF cannot provide.

Method	Micro-F1	Macro-F1	Weighted-F1	Subset Acc.
TF-IDF + LinearSVC (2k)	0.701	0.706	0.702	0.557
TF-IDF + LinearSVC (4k)	0.719	0.719	0.719	0.630
SDG Lens (MiniLM, 2k)	0.661	0.642	0.657	0.657
SDG Lens (MiniLM, 4k)	0.705	0.687	0.705	0.667

Table 1: Comparison of TF-IDF baseline vs. SDG Lens with varying train sizes (300-test split).

4.3 Per-Label Performance

Figure 2 shows that performance varies across SDGs. This is expected in a multi-label setting where classes differ in frequency and textual specificity. Several goals, including SDG 4, SDG 6, SDG 10, SDG 11, SDG 15, and SDG 17, reach comparatively strong F1 scores on the held-out test subset in this prototype run.

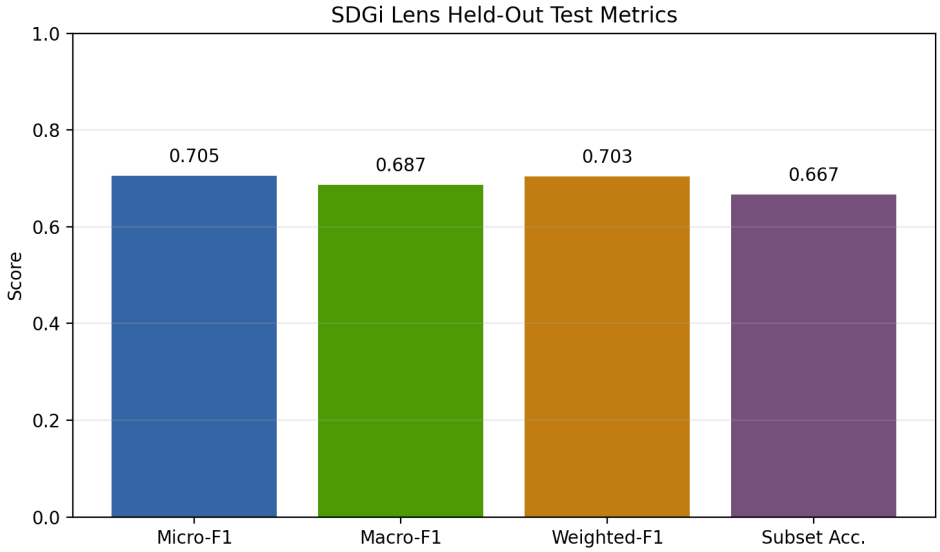

Figure 1: Out-of-sample classification metrics for SDG Lens on the held-out SDGi test subset.

4.4 Explanation Examples

Figure 3 presents five held-out test predictions: three correct cases, one partial multi-label match, and one incorrect case. The attention tokens often align with recognizable SDG concepts. For example, SDG 9 examples highlight terms such as "infrastructure", "innovation", and "industry", while SDG 11 examples highlight terms such as "housing", "urban", and "cities". This suggests that the model is attending to semantically relevant tokens, although attention should be interpreted cautiously.

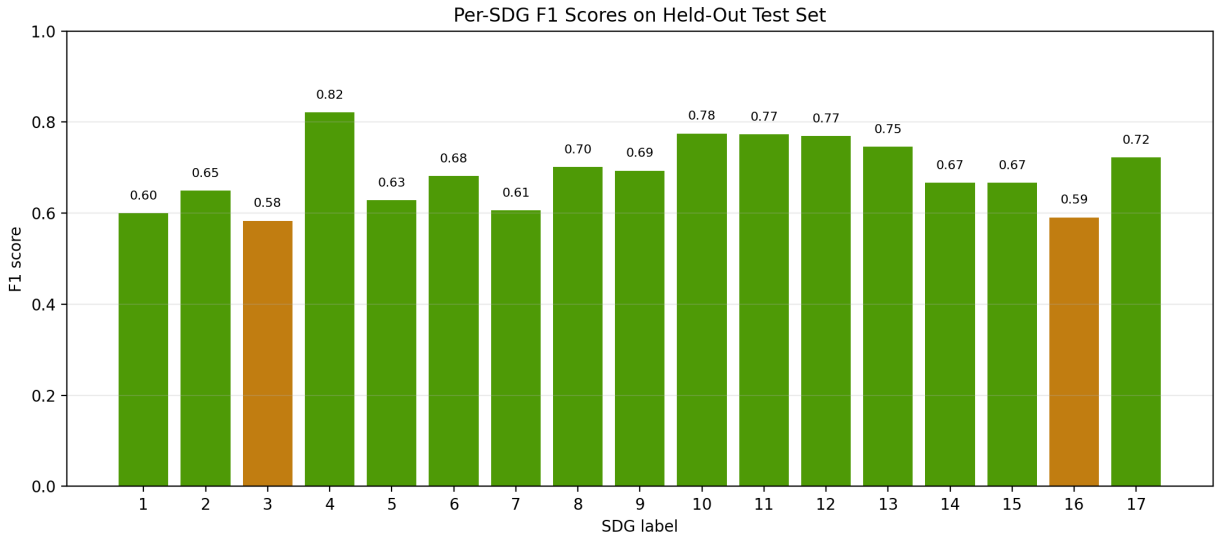

Figure 2: Per-SDG out-of-sample F1 scores for labels 1–17.

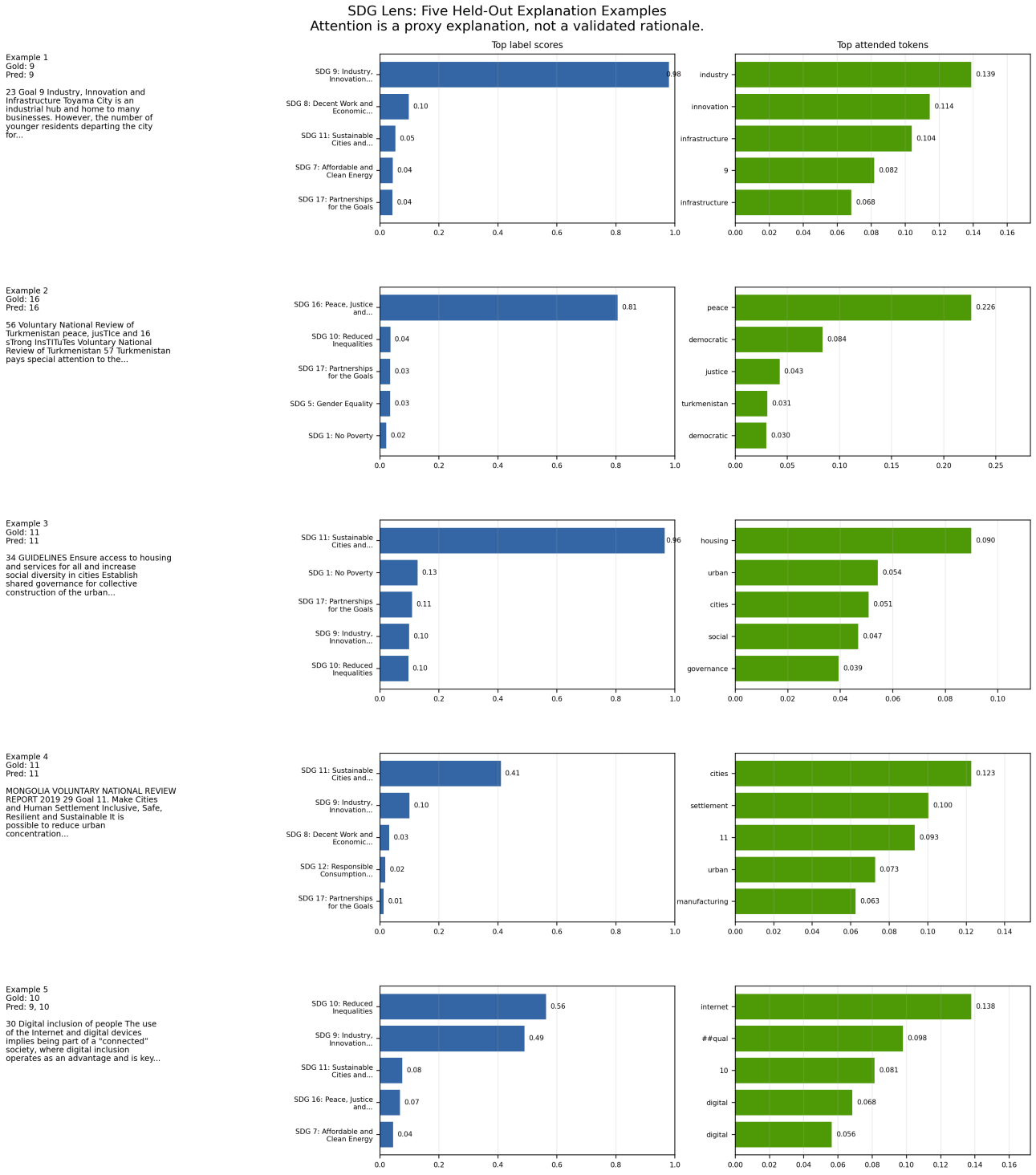

Figure 3: Five attention-based explanation proxy examples, including correct, partial, and incorrect predictions.

5 Discussion

The results indicate that explainable SDG classification is feasible with a small BERT-family model and a simple multi-label head. The attention outputs are especially useful as a reporting aid: they expose candidate terms that may have influenced the model's decision and make predictions easier to inspect. The multi-label framing is also appropriate because real SDG passages often overlap across goals, such as infrastructure and cities, or digital inclusion and inequality.

At the same time, the prototype remains limited. Attention is not equivalent to causal explanation, and the quality of highlighted tokens cannot be scored directly without token-level rationales. Threshold selection also affects the number of predicted SDGs, and different thresholds may be appropriate for precision-focused or recall-focused applications.

6 Limitations

First, SDGi labels are document-level labels, so supervision is weak for token-level explanation. Second, the explanation method uses attention as a proxy only; it does not prove causal influence. Third, results are from a prototype-scale English subset rather than full multilingual training. Fourth, the prediction threshold is fixed at 0.3 and was not tuned on a dedicated validation set. Finally, class imbalance and SDG overlap remain important sources of error.

7 Conclusion

SDG Lens demonstrates a compact route toward explainable multi-label SDG text classification. By combining a BERT-family classifier with attention-based token inspection, the system produces both SDG predictions and interpretable evidence signals. The prototype achieves micro-F1 around 0.66 and macro-F1 around 0.64 on a small English SDGi evaluation subset. Future work should evaluate explanations against human rationales, tune thresholds on validation data, and extend training to the full multilingual SDGi Corpus.

References

[1] United Nations Development Programme. *SDGi Corpus*. HuggingFace dataset: `UNDP/sdgi-corpus`.

[2] HateXplain authors. *HateXplain: A benchmark dataset for explainable hate speech detection*. Used here as motivation for attention-based token-level inspection.